

维度的边界 | 物理实在、感知投影与人类认知的空间划分

The Boundary of Dimensions: Physical Reality, Perceptual Projection and Spatial Division of Human Cognition

—— 角动量网络几何学 (ANG-TOE) 的维度分层定理

—The Dimension Stratification Theorem within Angular-Momentum Network Geometry (ANG-TOE)

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摘要 / Abstract

中文

本文在角动量网络几何学 (ANG-TOE) 框架中，提出一个根本性的维度划分定理：

我们严格证明：六维角动量旋转空间 $(\mathbf{L}, \mathbf{S})$ 是宇宙所有物理属性的完备基底，Axiom 0 ($\mathbf{J}_{\text{total}} \equiv 0$) 在此空间中运行。时频相空间 $(x, y, z, t, \omega, \Phi)$ 只是人类为了观测该物理本体而构建的读数界面，其所有坐标都依赖于局域拓扑密度 Φ ，因而不是独立的物理维度。

高于 6D 的数学结构 —— 希尔伯特空间、AI 参数空间、弦论高维、人类想象力 —— 全部是这两个空间在人类认知中的投影展开，而非宇宙的物理维度。

关键词：维度分层、角动量网络几何学、物理实在、感知投影、认知空间、ANG-TOE

English

Within the framework of Angular-Momentum Network Geometry (ANG-TOE), this paper proposes a fundamental theorem for dimension classification:

We rigorously demonstrate that the 6-dimensional angular-momentum rotation space $(\mathbf{L}, \mathbf{S})$ constitutes the complete basis for all cosmic physical properties, where Axiom 0 ($\mathbf{J}_{\text{total}} \equiv 0$) operates. The time-frequency phase space $(x, y, z, t, \omega, \Phi)$ is merely a readout interface constructed by humans to observe the physical ontology; all its coordinates depend on the local topological density Φ and therefore do not represent independent physical dimensions.

Higher-than-6D mathematical structures, including Hilbert space, AI parameter space, extra dimensions in string theory and human imagination, are projective expansions of the two fundamental spaces inside human cognition, rather than intrinsic physical dimensions of the universe.

Keywords: Dimension Stratification, Angular-Momentum Network Geometry, Physical Reality, Perceptual Projection, Cognitive Space, ANG-TOE

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1. 引言：维度混乱的根源 / 1. Introduction: The Origin of Dimension Confusion

中文

物理学在维度问题上面临一个长久未解的困惑：

- 人类日常感知：3 维空间 + 1 维时间 = 4D
- 广义相对论：在 4D 时空中运行
- 量子力学：使用无限维希尔伯特空间
- 弦论：需要 10D（超弦）或 26D（玻色弦）
- 人类想象力：似乎可以“进入”任意高维空间

这些“维度”之间是什么关系？弦论的 10D 是“物理”的吗？量子力学的无限维是“真实”的吗？人类的想象力是否在“进入”更高的维度？

本文的核心论点是：物理学的维度混乱，源于未能严格区分物理本体空间和人类感知投影空间。

我们提出一个清晰的维度分层：

层级	空间名称	坐标	物理地位
物理本体	六维角动量旋转空间	(L, S)	☒ 宇宙实在闭包
感知投影	时频相空间	(<i>x, y, z, t, ω, Φ</i>)	☒ 人类读数界面
认知展开	数学 / 想象空间	任意高维	☒ 认知投影

English

Physics has long faced an unsolved puzzle concerning dimensions:

- Daily human perception: 3-dimensional space + 1-dimensional time = 4D
- General Relativity: formulated in 4D spacetime
- Quantum Mechanics: employs infinite-dimensional Hilbert space
- String Theory: requires 10D (superstring) or 26D (bosonic string)
- Human imagination: appears capable of “accessing” arbitrarily higher-dimensional spaces

What is the relationship among these different “dimensions”? Are the 10 dimensions in string theory physically real? Is the infinite-dimensional space of quantum mechanics an objective feature of nature? Does human imagination truly reach higher physical dimensions?

The core argument of this paper is that confusion over dimensions in physics arises from the failure to strictly separate **the space of physical ontology** from **the space of human perceptual projection**.

We propose a clear stratification of dimensions:

Layer	Space Name	Coordinates	Physical Status
Physical Ontology	6D Angular-Momentum Rotation Space	(L , S)	☒ Closure of Cosmic Reality
Perceptual Projection	Time-Frequency Phase Space	(x, y, z, t, ω, Φ)	☒ Human Readout Interface
Cognitive Expansion	Mathematical / Imaginary Space	Arbitrary higher dimensions	☒ Cognitive Projection

2. 物理本体：六维角动量旋转空间 / 2. Physical Ontology: 6D Angular-Momentum Rotation Space

2.1 定义 / 2.1 Definition

中文

其中：

- **L**：轨道角动量（3 个独立方向分量）
- **S**：自旋 / 内禀角动量（3 个独立方向分量）

English

Where:

- **L**: orbital angular momentum (3 independent directional components)
- **S**: spin / intrinsic angular momentum (3 independent directional components)

2.2 为什么是“物理实在”？ / 2.2 Why It Represents “Physical Reality”

中文

在 ANG-TOE 中，链接网络是宇宙唯一的基本实体。每个链接由四个属性定义：

链接 = $(\mathbf{J}, \ell, \Theta, \Phi)$

其中 $\mathbf{J} = \mathbf{L} + \mathbf{S}$ 是链接的角动量通量，是唯一的“物理量”本体。

Axiom 0（最高宪法）：

$\mathbf{J}_{\text{total}} \equiv 0$

这个公理作用于六维角动量旋转空间，要求所有链接的角动量必须在全局边界上闭合。这正是“物理实在”的闭包条件——在六维角动量旋转空间中，宇宙的物理属性被完全确定。

English

Within ANG-TOE, the link network is the only fundamental entity of the universe.

Each link is defined by four properties:

Link = $(\mathbf{J}, \ell, \Theta, \Phi)$

where $\mathbf{J} = \mathbf{L} + \mathbf{S}$ denotes the angular-momentum flux of the link, the ontological origin of all physical quantities.

Axiom 0 (Fundamental Cosmic Axiom):

$\mathbf{J}_{\text{total}} \equiv 0$

This axiom acts inside the 6D angular-momentum rotation space and requires the angular momentum of all links to close at the global boundary. This closure condition defines physical reality: all cosmic physical properties are fully determined within this 6-dimensional manifold.

2.3 六维角动量旋转空间中的物理量 / 2.3 Physical Quantities inside the 6D Angular-Momentum Rotation Space

中文

物理量	在 $\mathcal{M}_6^{\text{物理}}$ 中的表达
质量	链接闭合网络的曲率能量： $m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi} \cdot \mathcal{F}$
力	角动量通量梯度： $\mathbf{F} = -\nabla \mathbf{J}$
能量	角动量时间变化率： $E = \partial \mathbf{J} / \partial t$
时间	重联事件速率： $\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$

在六维角动量旋转空间中，不存在“独立的”时间、空间或频率。它们都是角动量状态的投影读数。

English

Physical Quantity	Expression in $\mathcal{M}_6^{\text{Physical}}$
Mass	Curvature energy of closed link network: $m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi} \cdot \mathcal{F}$
Force	Gradient of angular-momentum flux: $\mathbf{F} = -\nabla \mathbf{J}$
Energy	Time rate of change of angular momentum: $E = \partial \mathbf{J} / \partial t$
Time	Rate of topological-reconnection events: $\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$

Within the 6D angular-momentum rotation space, there are no independent time, space or frequency dimensions; these quantities are merely projective readouts of angular-momentum states.

3. 感知投影：时频相空间 / 3. Perceptual Projection:
Time-Frequency Phase Space

3.1 定义 / 3.1 Definition

中文

这是人类用仪器定义的可测量坐标空间。

English

This is the measurable coordinate space defined by human instruments.

3.2 为什么不是物理本体？ / 3.2 Why It Is Not Physical
Ontology

中文

坐标	定义方式	物理依赖性
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t (秒)	铯-133 原子跃迁频率	依赖于局域 Φ (原子钟修正)
x, y, z (米)	光在 1/299792458 秒内传播距离	依赖于局域光速 $c(\Phi)$
ω (赫兹)	事件重复速率的测量值	依赖于局域重联速率
Φ (相位)	信号相位的读数	依赖于链接网络相干度

所有“感知维度”都依赖于局域拓扑密度 Φ 和局域光速 $c(\Phi)$ 。它们不是宇宙的常数，是人类测量工具在特定局域环境下的读数。

English

Coordinate	Definition	Physical Dependence
t (second)	Transition frequency of Caesium-133 atoms	Depends on local Φ (atomic clock corrections)
x, y, z (metre)	Distance travelled by light in 1/299792458 seconds	Depends on local light-speed $c(\Phi)$
ω (hertz)	Measured repetition rate of events	Depends on local reconnection rate
Φ (phase)	Readout of signal phase	Depends on coherence of the link network

All “perceptual dimensions” depend on the local topological density Φ and the local light-speed $c(\Phi)$. They are not universal cosmic constants but readouts generated by human measuring devices under local environmental conditions.

3.3 时频相空间中的物理量 / 3.3 Physical Quantities in Time-Frequency Phase Space

中文

人类感知的全部“物理量”，都是六维角动量旋转空间中角动量状态在时频相空间中的读数翻译。

人类感知量	在 $\mathcal{M}_6^{\text{感知}}$ 中的表达	对应 $\mathcal{M}_6^{\text{物理}}$ 中的量
速度	$v = \frac{dx}{dt}$	$\frac{\partial \mathbf{J}}{\partial \ell}$ (角动量梯度)
动量	$p = m \frac{dx}{dt}$	$\mathbf{J}/\ell$ (角动量除以半径)
能量	$E = \hbar\omega$	$\partial \mathbf{J} / \partial t$
作用量	$\mathcal{S} = \int E, dt$	$\oint \mathbf{J} \cdot d\mathbf{x}$

English

All “physical quantities” perceived by humans are translated readouts of angular-momentum states from the 6D angular-momentum rotation space onto the time-frequency phase space.

Human Observable	Expression in $\mathcal{M}_6^{\text{Perceptual}}$	Correspondence in $\mathcal{M}_6^{\text{Physical}}$
Velocity	$v = \frac{dx}{dt}$	$\frac{\partial \mathbf{J}}{\partial \ell}$ (angular-momentum gradient)
Momentum	$p = m \frac{dx}{dt}$	$\mathbf{J}/\ell$ (angular momentum divided by length scale)
Energy	$E = \hbar\omega$	$\partial \mathbf{J} / \partial t$
Action	$\mathcal{S} = \int E, dt$	$\oint \mathbf{J} \cdot d\mathbf{x}$

4. 严格区分：为什么这两个六维空间不等价 / 4. Strict Distinction: Why the Two 6-Dimensional Spaces Are Not Equivalent

4.1 核心差异 / 4.1 Core Differences

中文

对比维度	物理本体 $\mathcal{M}_6^{\text{物理}}$	感知投影 $\mathcal{M}_6^{\text{感知}}$
坐标	(L , S)	(x, y, z, t, ω, Φ)
物理地位	☒ 宇宙本体	☒ 人类读数界面
是否随局域 Φ 变化?	☒ 角动量为绝对量	☒ 长度、时间、频率随 Φ 改变
能否闭包?	☒ Axiom 0 全局闭包	☒ 仅为观测翻译层
数学角色	物理定律的运算基底	人类观测的翻译界面
因果关系	链接网络在此空间演化	将 6D 角动量状态翻译为可感知单位

English

Comparison Item	Physical Ontology $\mathcal{M}_6^{\text{Physical}}$	Perceptual Projection $\mathcal{M}_6^{\text{Perceptual}}$
Coordinates	(L , S)	(x, y, z, t, ω, Φ)
Physical Status	☒ Cosmic Ontology	☒ Human Readout Interface
Varies with local Φ ?	☒ Angular momentum is absolute	☒ Length, time, frequency change with Φ
Closure Property	☒ Globally closed under Axiom 0	☒ Merely an observational translation layer
Mathematical Role	Computational basis for physical laws	Translation interface for human measurement
Causality	Link network evolves within this manifold	Translates 6D angular-momentum

		states into measurable human units
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4.2 关键证明：为什么时频相空间不是物理空间？ / 4.2 Key Proof: Why Time-Frequency Phase Space Is Not the Fundamental Physical Space

中文

- 局域依赖性: $c(\Phi) = c_0 \cdot \sqrt{1 - \Phi/0.85}$, 光速随局域 Φ 变化, 因此“秒”和“米”是局域量。
- 原子钟修正: $\nu_{\text{true}} = \nu_{\text{Cs}} \cdot \mathcal{F}_\Phi \cdot \mathcal{F}_G \cdot \mathcal{F}_v$, 时间读数随引力势和速度变化。
- Axiom 0 不作用于时频相空间: 时频相空间不满足全局角动量归零的闭包条件。

English

- Local dependence: $c(\Phi) = c_0 \cdot \sqrt{1 - \Phi/0.85}$; light speed varies with local Φ , so the “second” and “metre” are local quantities.
- Atomic-clock correction: $\nu_{\text{true}} = \nu_{\text{Cs}} \cdot \mathcal{F}_\Phi \cdot \mathcal{F}_G \cdot \mathcal{F}_v$; measured time shifts with gravitational potential and velocity.
- Axiom 0 does not apply to time-frequency phase space: this perceptual manifold fails to satisfy the global closure condition of zero total angular momentum.

4.3 投射关系 / 4.3 Projection Relation

中文

人类在六维角动量旋转空间中从未直接观测。我们所有物理实验的“数据”都是在时频相空间中读取的读数。物理学真正的任务, 是把这些读数“翻译”回六维角动量旋转空间的几何结构。

English

Humans never directly observe the 6D angular-momentum rotation space itself. All experimental data are readouts obtained inside the time-frequency phase space. The true task of physics is to translate these measured readouts back into the geometric structures of the fundamental 6D manifold.

5. 六维之上：数学与认知的投影展开 / 5. Beyond Six Dimensions: Projective Expansion of Mathematics and Cognition

5.1 高于 6D 的空间是什么？ / 5.1 What Are Spaces with More Than Six Dimensions?

中文

高于 6D 的空间	ANG-TOE 中的解释
希尔伯特空间（无限维）	6D 角动量流形谱密度 $\mathcal{S}(\lambda)$ 的傅里叶展开。维度数量取决于基函数选择，不是物理实体。
AI 参数空间（亿级维）	6D 流形非局域耦合 Link_{ij} 在训练数据上的投影采样。本质上是查询表，不是物理空间。
人类想象空间（无限维）	CTL 网络在相域 (Φ) 中递归构造的虚拟链接路径。不消耗物理链接，只占用相位相干性。
弦论的 10D/26D	$\mathcal{M}_6^{\text{物理}}$ 在不同紧致化方案下的投影展开，非物理基底。

English

Higher-Dimensional Space	Interpretation in ANG-TOE
Hilbert Space (infinite-dimensional)	Fourier expansion of the spectral density $\mathcal{S}(\lambda)$ on the 6D angular-momentum manifold. Dimensionality depends on basis-function selection, not an objective physical entity.
AI Parameter Space (hundreds-of-millions dimensions)	Projected sampling of non-local couplings Link_{ij} of the 6D manifold over training data; essentially a lookup table rather than physical space.
Human Imaginative Space (infinite-dimensional)	Recursively constructed virtual link paths of the CTL network within the phase domain Φ . No

	physical links are consumed; only phase coherence is occupied.
10D / 26D in String Theory	Projective expansions of $\mathcal{M}_6^{\text{Physical}}$ under different compactification schemes; not the fundamental physical basis.

5.2 为什么高于 6D 不是“物理空间”？ / 5.2 Why Dimensions above 6 Are Not Physical Space

中文

- 无法被 Axiom 0 闭包：高于 6D 的空间无法定义全局角动量归零。
- 无法被链接网络承载：链接只有 4 个属性 $(\mathbf{J}, \ell, \Theta, \Phi)$ ，没有第 7 个属性。
- 无法被 T1-T17 覆盖：所有定理都指向 $(\chi, \beta_1, \text{Link})$ ，没有“额外维度”的对应量。

English

- Cannot be closed by Axiom 0: global zero total angular momentum cannot be defined for spaces beyond six dimensions.
- Cannot be carried by the link network: each link only has four attributes $(\mathbf{J}, \ell, \Theta, \Phi)$; there exists no seventh intrinsic property.
- Not covered by Theorems T1-T17: all core theorems refer to $(\chi, \beta_1, \text{Link})$, with no corresponding variables for extra physical dimensions.

5.3 数学工具的本质定位 / 5.3 Essential Nature of Mathematical Tools

中文

数学工具	是否高于 6D?	在 ANG-TOE 中的本质
虚数单位 i	—	Φ 的 90° 旋转投影算子 ($i^2 = -1$ 等价于 $\Phi \rightarrow \Phi + \pi$)
希尔伯特空间	☒ 无限维	6D 谱密度 $\mathcal{S}(\lambda)$ 在复平面上的展开基底

张量网络	☒ 高维	6D 流形集体重联行为的代数模拟器
范畴论 / 拓扑斯	☒ 任意高维	人类认知对 6D 拓扑结构的抽象化表达
人类想象力	☒ 无限维（可扩展）	CTL 网络在相域中的虚拟链接自指构造

人类之所以能想象高于 6D 的空间，不是因为宇宙本身有那么多维度，而是因为人脑的 CTL 网络可以在相域 Φ 中无限闭合虚拟链接，从而生成任意维度的认知投影。

English

Mathematical Tool	>6D?	Essence within ANG-TOE
Imaginary unit i	—	90°-rotation projection operator for Φ ($i^2 = -1$ corresponds to $\Phi \rightarrow \Phi + \pi$)
Hilbert Space	☒ Infinite-D	Expansion basis of the 6D spectral density $\mathcal{S}(\lambda)$ over the complex plane
Tensor Networks	☒ Higher-D	Algebraic simulator for collective reconnection behaviours on the 6D manifold
Category Theory / Topos Theory	☒ Arbitrary higher dimensions	Abstract human cognitive representation of 6D topological structures
Human Imagination	☒ Infinite expandable dimensions	Self-referential construction of virtual links by the CTL network inside the phase domain

Humans are able to imagine spaces with more than six dimensions not because the universe possesses those physical dimensions, but because the human CTL network can generate infinitely many closed virtual links within the phase domain Φ , producing arbitrarily-dimensional cognitive projections.

6. 对现有物理理论的重新定位 / 6. Repositioning of Existing Physical Theories

中文

物理理论	在 ANG-TOE 中的重新定位
弦论（10D/26D）	$\mathcal{M}_6^{\text{物理}}$ 在不同投影条件下的展开，不是额外物理维度。弦论的“紧致化”是时频相空间的读数校准，不是宇宙的物理事实。
量子力学（无限维希尔伯特空间）	6D 角动量流形谱密度在复平面上的展开基底。无限维是数学工具的便利，不是宇宙的物理属性。
广义相对论（4D 时空）	4D 时空是 $\mathcal{M}_6^{\text{感知}}$ 的子集（不含频率和相位）。GR 在低速弱场极限下有效，但不是宇宙的底层语法。
标准模型	粒子谱是 6D 角动量流形闭合拓扑结构的分类。所有量子数都是 $(\chi, \beta_1, \text{Link})$ 的投影读数。

English

Physical Theory	Repositioning under ANG-TOE
String Theory (10D/26D)	Expansions of $\mathcal{M}_6^{\text{Physical}}$ under different projection conditions, not extra physical dimensions. “Compactification” in string theory is readout calibration within time-frequency phase space rather than an objective cosmic feature.
Quantum Mechanics (infinite-D Hilbert space)	Expansion basis for the spectral density of the 6D angular-momentum manifold on the complex plane. Infinite dimensionality is a convenient mathematical tool, not an intrinsic property of the universe.
General Relativity (4D spacetime)	4D spacetime is a subset of $\mathcal{M}_6^{\text{Perceptual}}$ (excluding frequency and phase). GR is valid in the low-velocity weak-field limit but does not represent the fundamental grammar of the cosmos.
Standard Model	Particle spectra are classifications of closed topological structures on the 6D angular-momentum manifold. All quantum numbers are projective readouts of $(\chi, \beta_1, \text{Link})$.

7. 可证伪预测 / 7. Falsifiable Predictions

中文

- 1. 如果实验发现一个物理量无法用 6D 角动量旋转空间描述，则 ANG-TOE 的 6D 闭包被证伪。
- 2. 如果弦论在高于 6D 的物理实验中被直接验证（如观察到紧致化维度的信号），则 ANG-TOE 对高维空间的定位需要修正。
- 3. 如果人类想象力被证明无法超越 6D 流形的拓扑约束（即所有想象都严格被 6D 物理限制），则本文的“相域虚拟链接”假说被证伪。
- 4. 如果实验证明时频相空间的坐标不依赖于局域 Φ （即光速、时间、长度是绝对常数），则 ANG-TOE 的感知投影假说被证伪。

English

- 1. If an experimentally-measured physical quantity cannot be described within the 6D angular-momentum rotation space, the 6D closure postulate of ANG-TOE is falsified.
- 2. If string theory is directly verified via physical experiments for dimensions above six (e.g. detecting signals from compactified extra dimensions), ANG-TOE's interpretation of higher-dimensional spaces requires revision.
- 3. If human imagination is proven incapable of exceeding the topological constraints of the 6D manifold (all imagination is strictly bounded by 6D physics), the "virtual link in phase-domain" hypothesis of this paper is falsified.
- 4. If experiments demonstrate that coordinates inside the time-frequency phase space are independent of local Φ (light speed, time and length are absolute constants), the perceptual-projection hypothesis of ANG-TOE is falsified.

8. 结论 / 8. Conclusion

中文

区域	空间名称	坐标	物理地位
物理本体	六维角动量旋转空间	(L , S)	☒ 宇宙实在闭包

感知投影	时频相空间	$(x, y, z, t, \omega, \Phi)$	✗ 人类读数界面
认知展开	数学 / 想象空间	任意高维	✗ 认知投影

宇宙只有一个物理空间：六维角动量旋转空间。人类所有的“物理量”都是这个空间在不同投影层上的读数。时频相空间是读数面板，不是宇宙的底层语法。

English

Domain	Space Name	Coordinates	Physical Status
Physical Ontology	6D Angular-Momentum Rotation Space	$(\mathbf{L}, \mathbf{S})$	☒ Closure of Cosmic Reality
Perceptual Projection	Time-Frequency Phase Space	$(x, y, z, t, \omega, \Phi)$	✗ Human Readout Interface
Cognitive Expansion	Mathematical / Imaginary Space	Arbitrary higher dimensions	✗ Cognitive Projection

There exists only one fundamental physical space in the universe: the 6-dimensional angular-momentum rotation space. All human-measured “physical quantities” are readouts of this manifold projected onto different layers. The time-frequency phase space is merely a readout dashboard rather than the fundamental underlying grammar of the universe.

9. 参考文献 / 9. References

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